

Temperature, enzymes and metabolism in *E. coli*

What this model adds to what is published

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The question

- ▶ Metabolism is temperature-dependent; **thermal performance curves are the phenotype**
- ▶ An **etcGEM** predicts a whole TPC from *per-enzyme* thermodynamics — no curve fitting to the shape
- ▶ Three reference points, and every claim here says which one it is relative to:
 - ▶ **Li et al. 2021¹** — the only published etcGEM (yeast)
 - ▶ **Pettersen & Almaas 2023²** — re-ran it and found the inference unstable
 - ▶ **Madkaikar 2023³** — the MRes baseline this group inherited
- ▶ **What is new here: the data that makes the inference identifiable, and the audits that say what a fitted etcGEM does and does not know**

What an etcGEM is

- ▶ A genome-scale metabolic model of ~2 500 reactions, plus three things:
 - ▶ **an enzyme budget** — carrying flux costs protein, and protein is finite
 - ▶ **per-enzyme** $k_{cat}(T)$ — each enzyme's turnover peaks at its own T_{opt} and falls away either side (macromolecular rate theory, curvature $\Delta C_p^\ddagger$)⁴
 - ▶ **per-enzyme unfolding** — the folded fraction collapses at each enzyme's melting temperature T_m
- ▶ Warm the cell and every enzyme moves on its own curve; the LP re-allocates the budget
- ▶ **The organism's TPC is then emergent** — it is an output, not a fitted function

The two temperature responses, per enzyme

Left: turnover $\hat{k}(T)$, each enzyme peaking at its own T_{opt} . **Right:** folded fraction $f_N(T)$, each collapsing at its own T_m .

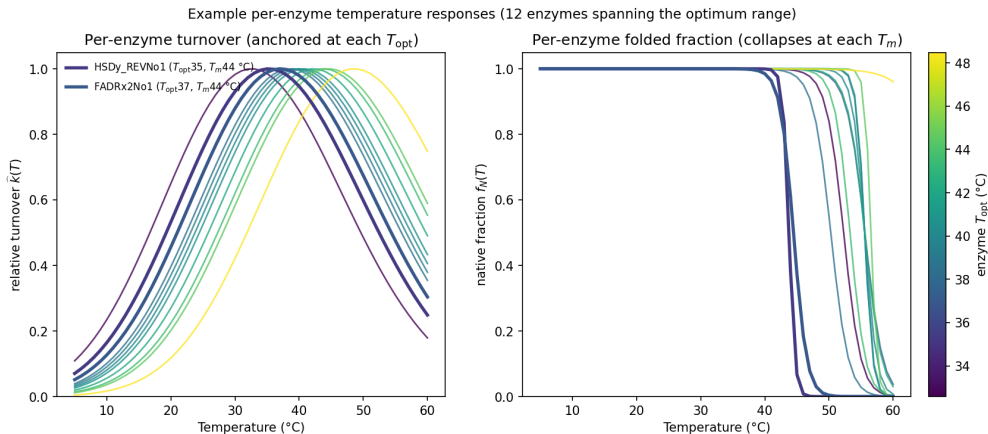


Figure 1: ~12 *E. coli* enzymes. [reports/ecoli_tpc](#)

The state of the art: Li et al. 2021

- ▶ Yeast ecYeast7; **764 enzymes, 2 292 parameters** (T_m , T_{opt} , $\Delta C_p^\ddagger$ each)
- ▶ T_m **measured** (266 of 764); T_{opt} **predicted** from sequence, carried as a wide prior
- ▶ Fitted to **growth and chemostat fluxes**
- ▶ **No gas exchange** — no O_2 , no acetate
- ▶ Headline: ERG1 controls flux at 40 °C, confirmed in the wet lab¹

What one of these curves looks like

Every descriptor a thermal biologist wants is read off **one predicted curve** — and each is a different question of the model.

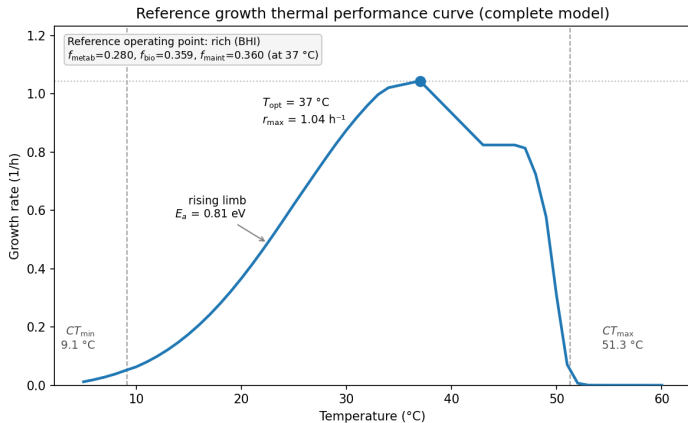


Figure 2: Our *E. coli* model at its reference point. `reports/ecoli_tpc`

The problem the field found: Pettersen & Almaas 2023

- ▶ “the Bayesian calculation method ... is **unstable and unable to estimate the posterior distribution**” — it assumes one peak; the problem has many
- ▶ Of six signature reactions, the one that varies most between **equally-fit** solutions is **ferrocytochrome-c:oxygen oxidoreductase** — “for some particles used extensively, in other cases not at all, still giving rise to approximately the same growth rates”
- ▶ Their diagnosis and their ask:
 - ▶ “the external measurements [are] insufficient to account for the inner workings of the ... cell”
 - ▶ “measurements of proteomics and **fluxomics** will help narrow down the solution space”²

The most under-determined reaction is the O₂-consuming step of respiration, and what they ask for is a measured flux. Hold that thought.

Where we started: the MRes baseline

- ▶ iJ01366, enzyme-constrained with GECKO 3 in MATLAB; **1 366 enzymes**
- ▶ **4 101 thermal parameters** — *more* per-enzyme freedom than Li et al.¹ — with T_m measured for 841 and T_{opt} from a database for 894
- ▶ Same thermal physics as now: two-state unfolding, transition-state k_{cat} with $\Delta C_p^\ddagger$, temperature-dependent maintenance
- ▶ **Not Bayesian**: repeated FBA scored by R^2 against **one** growth curve, the other 25 held out
- ▶ $R^2 = 0.94$; growth TPCs only³

Two things to be clear about: the GEM was later **replaced** (iJ01366 → eciML1515), and the baseline already had per-enzyme parameters — what it had no way to do was **constrain** them.

What the model is now

- ▶ One shared core in Python; **seven organisms** on it (*E. coli* best-constrained)
- ▶ **16 global levers**, deliberately — a manageable set whose identifiability we can test
- ▶ Everything is under a regression suite — **79 structural checks** and **60 numerical ones**, re-run on every change, so a fix to one organism cannot silently move another
- ▶ The figure is **a priori**: nothing was fitted to that curve

Emergent model vs Van Derlinden (K-12 MG1655, BHI; raw absolute rate, nothing fit)

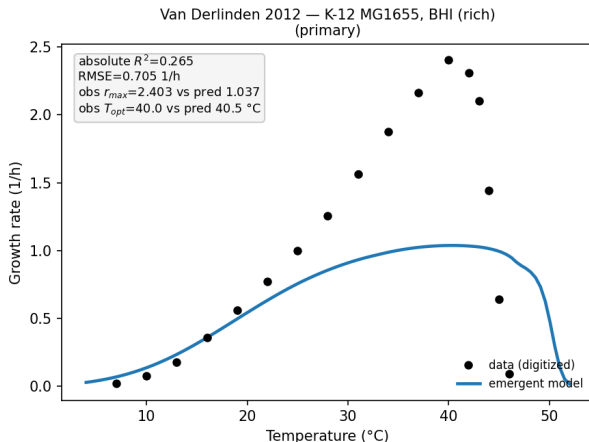


Figure 3: Van Derlinden et al.⁵, K-12 MG1655 in BHI.
reports/ecoli_tpc

The 16 levers — 1: the thermal envelope (6)

Each acts on **every** enzyme at once, on top of the per-enzyme data.

lever	what it does	what it is for
dT_{opt}	shifts every enzyme's T_{opt} by a constant	the database's optima are systematically off
topt_scale	stretches the <i>spread</i> of optima about 30 °C	how synchronised the enzymes are
ΔC_p _scale	scales the curvature of $k_{cat}(T)$	how broad each enzyme's response is
dT_m	shifts every melting temperature	<i>in vitro</i> $T_m \neq$ <i>in vivo</i> inactivation
tm_scale	stretches the <i>spread</i> of melting points	sharp collapse vs gradual shoulder
kcat_scale	scales the <i>level</i> of all turnover	the <i>in vitro</i> $\rightarrow$ <i>in vivo</i> k_{cat} gap ⁶

The 16 levers — 2: budget, maintenance, medium, observation (10)

lever	what it does
σ	average <i>in vivo</i> enzyme saturation — how much of the protein is working
kappa_scale	effective translation efficiency in the biosynthesis cap
f_{metab} , f_{maint}	the proteome split between flux-carrying and housekeeping sectors
ngam_scale, ngam_steepness	size of the maintenance cost, and how fast it rises with T
clearance_mult	how fast the medium is actually drawn down (a <i>medium</i> lever)
resp_scale	mmol O ₂ → measured units: an observation scale, not biology
disc_resp, disc_growth	model discrepancy — how much the model is allowed to be wrong

Which of the sixteen the data can actually move

Three kinds, and the difference matters — not two.

- ▶ **Eight the data move.** dT_{opt} , $topt_scale$, ΔC_p_scale , dT_m , tm_scale , $kcat_scale$, σ , $ngam_scale$
- ▶ **Two are FIXED BY MEASUREMENT** — f_{metab} , f_{maint} . They enter from the measured proteome, matched to medium *and* temperature, and carry **tight priors: measurement wiggle only**. The likelihood not moving them within that width is **the design working**, not a deficiency — growth and respiration are not asked to re-derive a proteomics measurement
- ▶ **Two are genuinely unidentified.** $kappa_scale$ (effective translation demand, derived once at build time — nothing measures it independently) and $ngam_steepness$ (maintenance steepness, *borrowed* from the MRes model). **These** are the ones a measurement would fix
- ▶ **Four are not model levers at all.** $clearance_mult$ is a property of the experiment; $resp_scale$, $disc_resp$, $disc_growth$ are the observation model

...and a fourth kind the categories above miss

- ▶ **Two stability levers can compensate for each other.** Likelihood profiles find similarly good growing fits with different combinations of dT_m and `tm_scale`
- ▶ **That differs from an unmeasured lever.** The fit constrains an effective low-temperature tail of the meltome more strongly than it distinguishes these two ways of moving it
- ▶ The failed 16-parameter run also had a thin direction involving this pair among its remaining sampling points
- ▶ **Those points describe the run's remaining high-likelihood region, not the full posterior.** Their spread cannot establish which parameters the posterior determines individually

Development 1: proteome allocation, from measured proteomes

- ▶ The enzyme budget is split into three **sectors**^{7,8}: **metabolic** (carries flux) · **biosynthesis** (ribosomes) · **maintenance** (chaperones, housekeeping)
- ▶ The split is **not fitted and not hand-set**. It is read from a study that *measured the E. coli proteome at a series of temperatures*, in separate LB / glucose / glycerol series⁹
- ▶ So the model is condition-specific for free: measured ribosome fraction at 37 °C is **0.36 on rich LB** but **0.19 on glucose-minimal** — a rich-medium run automatically gets a higher biosynthesis cap and grows faster
- ▶ The **maintenance** sector carries a measured **2.7-fold heat-stress chaperone rise**
- ▶ New relative to Li et al.¹ and to the MRes³: **neither has an allocation layer at all**

The measured sectors

The chaperone sector (red) ramps sharply above 37 °C — the measured heat-stress response entering as an allocation change, not a fitted parameter.

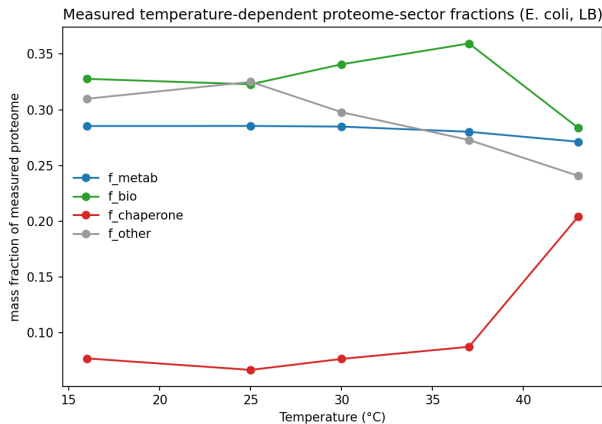


Figure 4: Wang et al.⁹; figure reports/ecoli_tpc

...and this is where we need more data

- ▶ **The measurement is thin where it matters most.** The glucose series was measured at **25, 30 and 37 °C only**; the LB series adds 16 and 43 °C
- ▶ Outside a series' range the allocation is **held at its endpoint** — so above 37 °C on glucose the sector split, and with it the translation cap, **stops varying with temperature**
- ▶ The consequence is visible in the curve: the glucose-minimal prediction is **bit-identical from 37 to 44 °C** (growth exactly 0.524757 h^{-1} at all eight grid points)
- ▶ **That shoulder is the proteomics table running out, not a property of the cell** — and it is why “ T_{opt} ” from such a run is the argmax of a tie

This is the concrete version of “we need more data”: extend the measured glucose series above 37 °C. It is a proteomics experiment, not a modelling problem.

Which lever owns which feature of the curve

- ▶ The heatmap shows **12 rows, not 16**: `clearance_mult` is a property of the experiment, and `resp_scale/disc_resp/disc_growth` are the observation model — none of the four touches the predicted curve
- ▶ Read the rows in four groups:
 - ▶ **catalysis** = $\sigma + \text{kcat_scale}$ — how much working enzyme there is
 - ▶ **kinetic envelope** = $dT_{opt} + \text{topt_scale} + \Delta C_p\text{-scale}$ — where the optima sit and how spread
 - ▶ **stability** = $dT_m + \text{tm_scale}$ — where unfolding takes over
 - ▶ **allocation / maintenance** = $f_{metab}, f_{maint}, \text{kappa_scale}, \text{ngam_}$
- ▶ In lever terms: $dT_{opt}/\text{topt_scale}$ move the **cold** limb and E_a ; $dT_m/\text{tm_scale}$ move the **hot** collapse; $\sigma/\text{kcat_scale}$ move the **height** and nothing else
- ▶ **The result**: r_{max} is owned by catalysis (σ alone has elasticity ≈ 1); T_{opt} and CT_{max} by stability; E_a and niche width by the spread of optima
- ▶ **So a TPC descriptor is not a parameter**. Each one asks the model a different question — which is why fitting to peak height tells you nothing about the hot edge

Elasticity: the numbers

Fractional change in each descriptor per fractional change in each lever, at the operating point. **Deterministic — no intervals.** Bold is $|E| > 0.3$.

lever	T_{opt}	r_{max}	CT_{max}	niche width	E_a
dT_{opt}	0.00	-0.09	0.00	-0.34	0.26
topt_scale	0.00	-0.23	0.00	-1.05	1.41
ΔC_p _scale	0.00	-0.02	0.00	0.29	0.22
dT_m	0.13	0.09	0.13	0.18	-0.24
tm_scale	-0.26	-0.14	-0.16	-0.21	0.24
kcat_scale	0.00	0.40	0.00	0.07	-0.37
σ	0.00	1.00	0.00	-0.11	0.36
f_{maint}	0.00	-0.56	0.00	0.06	-0.37
ngam_scale	0.00	-0.00	0.00	-0.01	0.01
the three zero rows	0.00	0.00	0.00	0.00	0.00

...and the same numbers as a pattern

The three zero rows are zero for **different reasons**: kappa_scale and ngam_steepness unidentified, f_{metab} fixed by measurement.

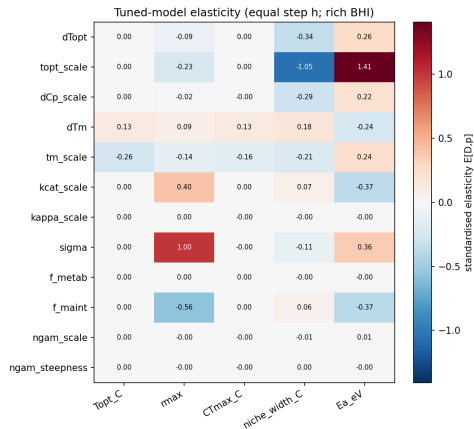


Figure 5: Elasticity heatmap, $h = 0.10$. reports/ecoli_tpc

Development 2: overflow metabolism — what it is

- ▶ Grow *E. coli* fast enough and it **excretes acetate while oxygen is still available** — it ferments and respire at once, wasting carbon. The bacterial analogue of the Crabtree effect
- ▶ Measured law: excretion is zero below a growth-rate threshold, then linear.
Threshold $\approx 0.76 \text{ h}^{-1}$, slope $\approx 21 \text{ mmol gDW}^{-1} \text{ h}^{-1}$
- ▶ **Why any of this matters for a temperature model:** an enzyme-constrained TPC fixes the growth rate but *not* the internal flux distribution, so **O₂ and CO₂ exchange are under-determined** — the model can hit the right growth rate by respiring or by fermenting
- ▶ That is the *same* degeneracy Pettersen & Almaas found by FVA on the oxidase reaction², seen from the other side
- ▶ **Until it is pinned, an etcGEM cannot predict a gas flux at all**

The total-carbon cap: what it is

- ▶ **From the code:** $\sum_i n_{C,i} v_i \leq c_{max}$ over **every open carbon uptake**, with $n_{C,i}$ the carbon count read from the model's own metabolite formula. Units **mmol C gDW⁻¹ h⁻¹**; inorganic carbon is not charged to it
- ▶ **What it produces:** a finite carbon budget *and* a finite protein budget make the LP excrete acetate on its own. **Overflow emerges; it is not written in.** That is the difference between configuration D and configuration C
- ▶ **A second role, honestly stated:** an enzyme-constrained curve fixes growth but not the internal fluxes, so O₂ and CO₂ are under-determined. The cap is **one of three ways offered to pin them** — and it does **not** fully succeed: the O₂ flux at the optimum still jumps between equally-optimal solutions, which is the cliff problem later in this deck

The total-carbon cap: why the value matters

- ▶ **At $c_{max} = 60$ acetate overflow is exactly zero on both media** — the cap suppresses the very mechanism it exists to create. On NLDM, T_{opt} also drops **39.0 → 32.5 °C** between 80 and 60
- ▶ **120 is the lowest value in his sweep at which overflow is non-zero on both** (23.5 on glucose, 3.9 on NLDM mmol gDW⁻¹ h⁻¹) — which is why it was adopted
- ▶ **It is medium-dependent, and 60 / 120 / 450 is one quantity, not three guesses.** Applying the glucose/NLDM value of 120 to LB holds r_{max} at **1.1–1.8 h⁻¹** against a measured **2.945**, and growth R^2 falls to **0.64 / 0.05 / -0.10** across three configurations; at the LB-appropriate cap they are **0.90 / 0.83 / 0.88**
- ▶ **LB is settled at 450.** That was the resolution of a real disagreement, not a compromise

...and how it is made to emerge

Four ways to pin it — Parsa's configurations A–F, ported here and reproduced against his numbers.

	mechanism	imposed or emergent
A	cost the membrane carriers: transporters get the same thermal envelope as every other enzyme	emergent ceiling
B	a total-carbon cap , $\sum_i n_{C,i} v_i \leq c_{max}$	emergent
C	the measured acetate line written in as a constraint ⁷	imposed
D	the carbon cap, oxidase split left free — the LP picks <i>bd</i> vs <i>bo3</i> per temperature	emergent, T-dependent

Why configuration D is the interesting one

- ▶ **Nothing is imposed on the acetate branch.** Overflow falls out of the protein budget plus a carbon budget — the model is not told that acetate appears above 0.76 h^{-1} , it produces it
- ▶ The **terminal-oxidase choice becomes a prediction:** *bd* and *bo3* have different melting temperatures, so which one the cell uses shifts with temperature on its own
- ▶ c_{max} is **not our number**: Parsa's own sweep (60 / 100 / 120 / 180 mmol C gDW⁻¹ h⁻¹) concludes 100–120 is the sweet spot — *“a tight cap gives respiratory gas but a flat growth plateau; a loose cap gives a peaked TPC but fermentative gas”*. 120 adopted; 450 on LB
- ▶ Data behind the layer: the **acetate law**⁷, the **membrane geometry** (Szenk et al., surface-to-volume $4.5 \mu\text{m}^{-1}$), and Parsa's own **respirometry** for the fits
- ▶ **The advance over Li et al.¹ and the MRes³: both stop at growth.** This turns O₂, CO₂ and acetate into predictions that a measurement can check — and Parsa's respirometry is that measurement

Configuration D: the growth curve it predicts

The medium ladder is the measured sector split at work: r_{max} **1.19** (glucose) to **2.02** h^{-1} (BHI/LB), T_{opt} 38–40 °C. **Model output; no measurement plotted.**

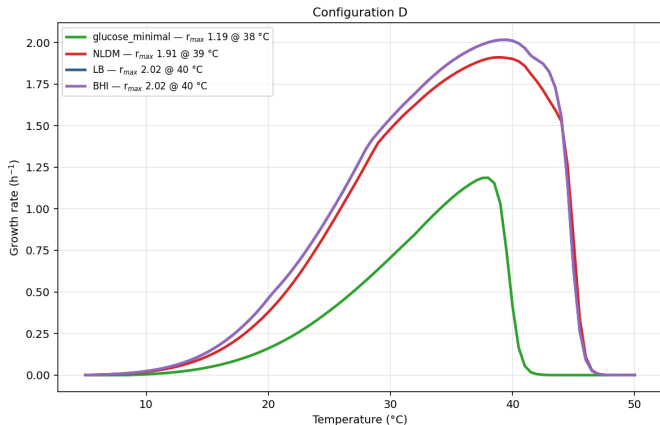


Figure 6: reports/ecoli_gasflux

...and the gas exchange it predicts

Right-hand panel: $RQ \approx 1$ around the optimum — respiratory, not fermentative.
Model output; no measurement plotted; mechanism, not interval.

Configuration D — gas exchange (collapsed tail masked)

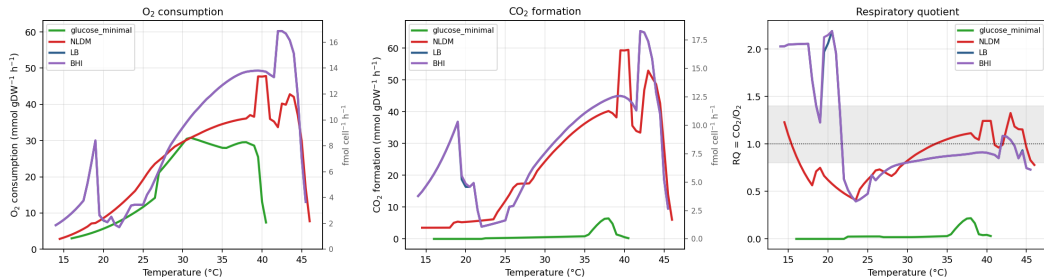


Figure 7: reports/ecoli_gasflux

The data nobody else has

- ▶ Parsa's respirometry: **O₂ flux and growth per cell, 15–50 °C, three media** (R2A, LB, M9), ~60 replicate runs each
- ▶ This is exactly the **fluxomics on the O₂-consuming reaction** that Pettersen & Almaas² name as the missing data
- ▶ And it is a **measurement**, not a model output — which is what makes it able to break the degeneracy
- ▶ Ported and **checked against his own results**: all sixty of his configuration numbers and all ten of his respirometry fits reproduce here, the worst to within 0.009
- ▶ Look at the shape: **respiration peaks 3–5 °C above growth** (43–45 °C vs ~40 °C) and is still climbing where growth has begun to fail

Measured growth and respiration, three media

Respiration peaks 3–5 °C above growth and is still climbing where growth has begun to fail. Faint points are replicate runs; bars s.e.m.

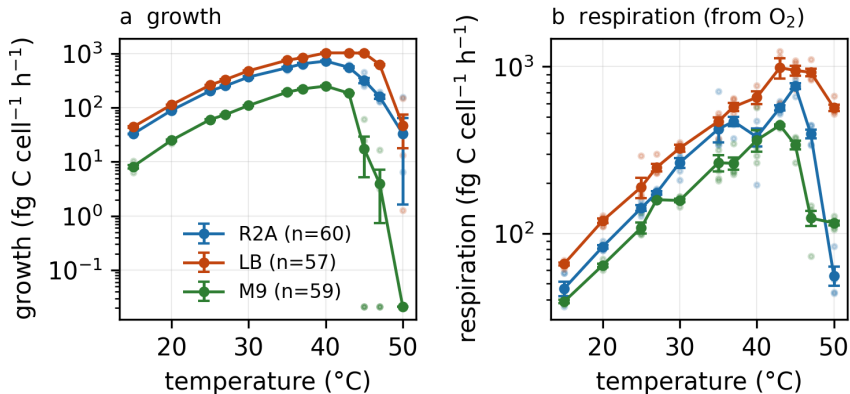


Figure 8: [strains/eciML1515/respirometry/](#)

How well does it fit? Growth

Lines are the model, points the measurement; two media, three configurations. **At Parsa's gated parameters — not a converged posterior, and no intervals.**

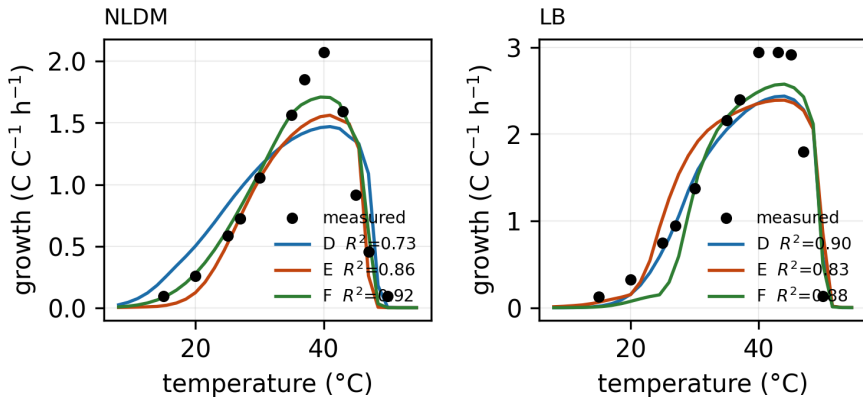


Figure 9: R^2 from reports/P3_gate/gate_def_headline.csv. make_fit_figures.py

How well does it fit? Oxygen

Gated parameters, no posterior. A fitted scale converts model to measurement: this tests **shape**, not level. **The jagged cold limb is the vertex-jumping from the cliff slides.**

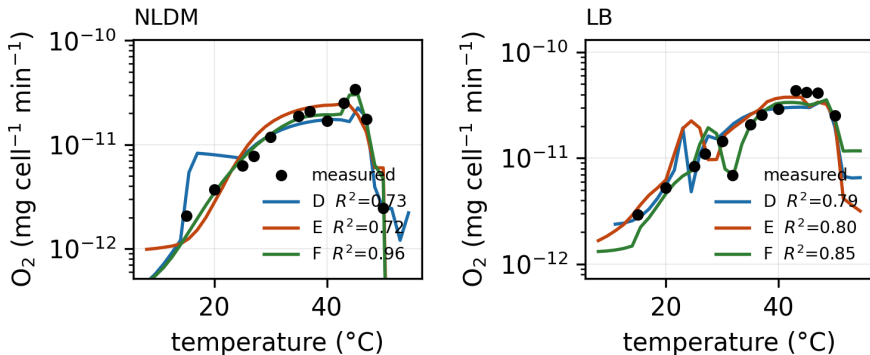


Figure 10: make_fit_figures.py

...and what it costs the cell

- ▶ **Carbon-use efficiency** — the share of assimilated carbon kept as biomass rather than respired
- ▶ Flat at **~0.6** (R2A, LB) and **~0.4** (M9) from 25 to 40 °C
- ▶ Then a **cliff**: 0.29 / 0.52 / 0.05 at 45 °C
- ▶ Medium sets the *level*; temperature sets the *cliff*
- ▶ **The model does not yet predict it.** CUE needs the carbon budget closed — biomass carbon against total carbon consumed — and that is the next thing the gas-exchange layer can be asked for

Measured carbon-use efficiency

The plotted conversion gives a warm-end decline. An unresolved carbon-per-cell constant changes both CUE levels and temperature dependence.

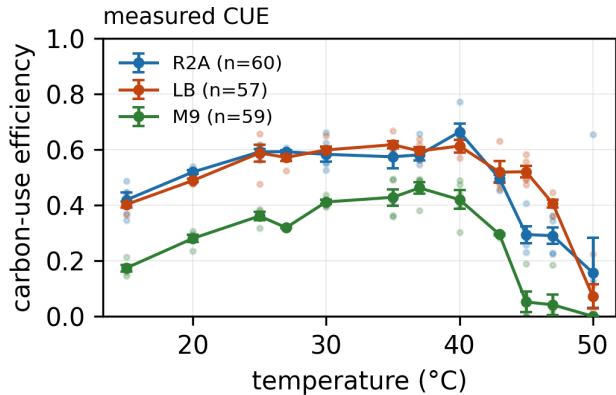


Figure 11: strains/eciML1515/respirometry/

...and carbon-use efficiency, which we cannot yet overlay

- ▶ The model would need **biomass carbon over total carbon consumed**
- ▶ The second term is a sum over the **64 open carbon sources** of the NLDM recipe, which the prediction path does not return; the first needs a **biomass carbon fraction** that is not in the strain data
- ▶ Both would have to be invented, so **CUE stays measurement-only** — the measured curve is the previous slide
- ▶ **This is the next thing to close**, and it is a small piece of plumbing rather than a new experiment

The statistical problem, in plain terms

What everyone is trying to do: the per-enzyme parameters are uncertain, so treat them as unknowns and ask the data which values are consistent with it — a *distribution* over parameters, not one best guess.

- ▶ **Li et al.** used **SMC-ABC**: draw parameter sets from the prior, simulate, keep the ones whose predictions land close to the data, narrow the tolerance, repeat. Reported a posterior over 2 292 parameters¹
- ▶ **Pettersen & Almaas** re-ran it **from different random seeds and got different answers**. The method assumes the good parameter sets form a single hill; in this problem there are many. They swapped in an **evolutionary algorithm** to collect a *diversity* of good solutions instead — then showed those solutions disagree wildly on internal fluxes²
- ▶ **Our inference problem:** 16 global parameters and a likelihood combining growth with **measured respiration**. We started with ensemble MCMC and moved to nested sampling after diagnosing the fitting surface

...and the three things that went wrong

1. **The chains were too short.** In the earlier MCMC-based family — ours and Parsa's — no chain reached the adopted target (effective sample size ≥ 600), and running longer did not help. **Intervals from those older chains are unsupported.** That is why the method changed to nested sampling
 2. **The surface itself is not smooth.** We stopped blaming the sampler and looked at the likelihood: 22 line scans, 902 model builds, **12 of 22 lines jump by >20 % of their range on a small step.** Not the solver — unchanged at 10^{-9} tolerances and on a different algorithm
 3. **The cause is the respiration term, and it is the degeneracy again.** At cold temperatures the O_2 flux at the growth optimum **jumps between equally-optimal solutions** — so the prediction moves discontinuously while the growth rate does not. *Exactly* what Pettersen & Almaas saw by FVA², here as cliffs in a likelihood
- **Three fixes, each for a named mechanism:** a **tie-break** to choose one point on the degenerate face; a **variance floor** so the likelihood is not shocked by a jump the model cannot avoid; a **continuous support** clamp at the growth threshold. Cliffs cut $\sim 10\times$. **Still not smooth, and we say so**

What a line scan is, and what it found

- ▶ **A transect through parameter space.** Start at the best fit, choose a parameter or combination, and step along it. Rebuild and solve the whole model at each point, recording fit quality
- ▶ **22 directions, 41 points each: 902 model builds.** Each step is 0.05 prior standard deviations. Each line spans -1 to $+1$ standard deviation
- ▶ **12 of 22 directions have a cliff.** One small step changes fit quality by more than one fifth of its total change along the whole line. Median largest jump: 39 % of that line's range, worst: 99 %
- ▶ Small parameter changes can therefore give abruptly worse fits and impede local searches. Two directions are flat. The next slide shows all 22 transects

The surface, scanned

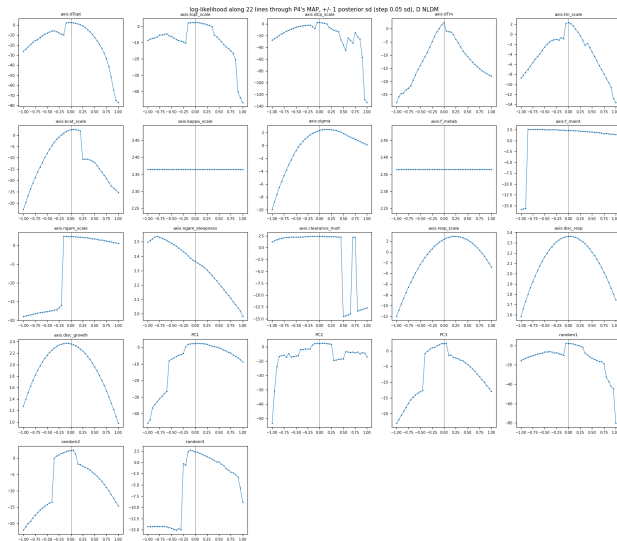


Figure 12: The log-likelihood along 22 parameter lines. reports/P9_surface

How many answers are there? First, what a basin is

- ▶ **A basin is a valley of the fitting problem.** Nearby starting parameters lead an optimiser towards the same good fit
- ▶ **A live basin predicts growth. A dead basin predicts essentially none**, although its parameters still receive a finite statistical score
- ▶ We screened **100 starts**, then continued 12 representative endpoints. **11 converged**. We scanned paths between endpoints to look for barriers in fit quality
- ▶ **The final rule was fixed before those results:** separate basins require a barrier above **5 log-likelihood units** and locally optimal endpoints. Barriers of 1–5 are substructure, below 1 is noise

How many answers are there? The result

- ▶ **The map found one live and one dead region.** The result holds at barrier thresholds of 3, 5 and 8 units. Straight-line paths give an upper bound on the number of basins
- ▶ The dead region predicts peak growth **0.000 h^{-1}** , against a measured **2.076 h^{-1}**
- ▶ **Eleven of its twelve temperatures are infeasible:** the metabolic constraints admit no flux solution. Those temperatures receive no respiration score
- ▶ **The dead region survives the current support rule.** The live–dead likelihood gap changed only from 11.69 to 11.25 units. Infeasibility remains exempted from respiration scoring

The reduced posterior: run 1 converged

- ▶ We fixed the uniform melting-temperature shift $dT_m = 0$, leaving **15 free parameters**. Configuration D on the NLDM recipe, with the current respiration likelihood
- ▶ **The first clean nested run in this family:** 14,088 iterations, 221,781 likelihood evaluations, **9.585 hours**, no crash
- ▶ **Log evidence** $\log Z = -26.030 \pm 0.109$, effective sample size **5,988**. The recorded stopping statistic reached **0.09999**, below 0.1
- ▶ At the previous run's failure point, **iteration 11,547**, dlogz was **0.666 versus 2.168**, at a comparable evaluation count
- ▶ **Seed 2 landed, and the two seeds did not agree.** Evidence agreed ($\Delta \log Z = 0.022$ against a combined error of 0.148) but **14 of 15 parameter medians differed by more than two Monte-Carlo errors**, so the pre-registered rule reads DISAGREE
- ▶ **This run has since been superseded.** Its likelihood exempted infeasible models from the respiration score; the revision that fixes that is two slides on

The stopping and reproducibility rules

- ▶ **Pre-registered sampler:** dynesty, 800 live points, random-direction slice sampling (rslice, 3 slices), multiple ellipsoidal bounds
- ▶ **Stopping rule:** estimated remaining change in log evidence, $\mathbf{dlogz} < 0.1$. Evidence uncertainty is a separate quantity
- ▶ **Agreement requires both:** the two log evidences agree within their combined reported errors **and** no parameter median differs by more than two Monte-Carlo errors
- ▶ **The test has a result: DISAGREE.** Both seeds met the stopping rule; the evidences agreed and the medians did not. Sampler convergence alone does not establish useful biological predictions — and here it did not even establish reproducibility
- ▶ **The same rule has since been applied twice more**, to the revised target (three slides on) and with the same verdict

What the reduction assumes

- ▶ **tm_scale does not reach its hard upper bound:** seed 1 median **1.069** [0.793, 1.515], seed 2 median **0.956** [0.783, 1.228], prior bounds **[0.4, 2.2]**; **0.00%** of either run's weight lies within 5% of 2.2
- ▶ **But under the revised likelihood it presses against the prior's *mass*:** medians **1.319** and **1.232**, sitting at the **99.4th percentile of the prior** in every run. The hard bound is never reached and the prior is still doing the constraining — the distinction matters and both numbers are given
- ▶ The failed 16-parameter run identified a numerical difficulty in its remaining sampling points. **That geometry cannot be treated as the full posterior**
- ▶ **Fixing dT_m assumes the meltome's mean is right.** Any uniform error in measured melting temperatures must now be absorbed by **tm_scale** and the catalytic parameters

Run 1 assigns about half its weight to very low growth

Measured peak: **2.0761 h⁻¹**. Results below are provisional, from seed 1.

summary	log likelihood	peak growth, h ⁻¹
Best posterior sample (MAP)	-10.704	1.7255
Separate median of each parameter	-30.558	0.0043

- ▶ The MAP reaches **83% of the measured peak**
- ▶ **300 draws sampled using posterior weights:** median peak **0.0016 h⁻¹**, 5th–95th percentiles **[0.0000, 1.7106]**
- ▶ **50.7% of these draws grow below half the measured peak.** This estimates low-growth posterior mass, not the fraction proven infeasible
- ▶ **This is the finding the target revision removed.** Under the revised likelihood the same quantity is **1.4% and 2.4%** in two independent runs: the low-growth mass was the scoring rule, not the biology

Why separate parameter medians give a misleading model

- ▶ Correlated parameters work in combinations. Taking each parameter's median separately can put the resulting vector far from combinations that jointly fit well
- ▶ Here that vector predicts peak growth **0.0043 h^{-1}** , against **1.7255 h^{-1}** at the best posterior sample
- ▶ Its growth $R^2 = -2.01$ describes this unrepresentative vector. It cannot summarise the predictive performance of the full posterior
- ▶ **Report the MAP alongside predictions drawn from the weighted joint posterior.** The median vector appears here only to show why it is unusable

The infeasibility exemption: diagnosed, then fixed

Earlier basin-map scores, before the current support rule:

log-likelihood contribution	live region	dead region
Growth	+6.363	−18.860
Respiration	−13.549	−0.017

- ▶ **The dead region gains about 13.5 units on respiration** because 11 of 12 temperatures have no feasible flux solution and receive no respiration score
- ▶ **The share was then measured, not guessed:** of the 800 sampling points retained by the run above, **765 are infeasible at every measured temperature** and carry **81.5% of its posterior weight** — from about **16%** of the prior's volume
- ▶ **The rule adopted:** a parameter set with no feasible flux solution at any measured temperature has **zero likelihood**. This is the limit of the existing log-scale term, not a penalty and not a fitted constant

Why a converged posterior can favour poor predictions

- ▶ The earlier live–dead gap is about **11.7 log-likelihood units**, a **dead-to-live likelihood ratio of roughly 8×10^{-6}**
- ▶ Posterior mass also depends on **how much prior volume each region occupies**. A sufficiently larger dead region can outweigh that fit disadvantage
- ▶ Illustration only: in 15 dimensions, a live region occupying **46% of prior width per axis** has volume about 8×10^{-6} of the full box. The actual regions need not be rectangular
- ▶ **The cause was confirmed and the rule changed.** Convergence did not validate the likelihood; the measurement did, and the scoring rule now excludes the dead region rather than rewarding it. This finding concerns this model and scoring rule, not enzyme-constrained models in general

The revision worked — and the runs still disagree

Configuration D on NLDM, 15 sampled parameters (one was never read by the model), infeasible sets scored at zero likelihood. Three independent runs on reserved seeds.

- ▶ **Every posterior sample is now feasible at every measured temperature**, and **97.6–98.6%** of predictive draws grow above half the measured peak. The dead region is gone, as designed
- ▶ **The prior rejects 16.5%** of itself as infeasible [14.9, 18.1], almost always at the coldest temperature — a property of the model worth explaining, not a defect of the fit
- ▶ **The two runs that finished still disagree:** $\log Z = -27.707 \pm 0.110$ against -28.047 ± 0.100 , and **all 14 physical medians** differ by more than two Monte-Carlo errors
- ▶ **A positive control failed too.** A coordinate that provably enters nothing should recover its own prior; in both runs it does not
- ▶ **The third run crashed.** Its cause is the next slide

The likelihood is not a function of its parameters

The crash point was stored as **−18.480**; re-evaluated from a clean process it is **−19.761**, every time — **1.28 too high**, and below the threshold it was meant to clear.

scheme	max deviation from a clean evaluation	cost
current	1.815	1.79 s
reset the solver each call	23.277	2.91 s
single-solve lexicographic	different vertex — disqualified	1.49 s
a fresh model each call	0.000	5.70 s

- ▶ The oxygen flux can sit on a *face* of the linear program, and which point the solver returns depends on what that worker solved before. **Only a fresh model per evaluation is exact**, at **3.2×** cost
- ▶ **This does not explain the disagreement above** — those samples re-evaluate clean

Auditing the published model

- ▶ We built an audit for a defect class we had found in our **own** models — a free ion path that lets ATP synthase run without the respiratory chain paying for it — and ran it on Li et al.'s deposited model¹
- ▶ **The defect is absent.** Their respiratory chain supplies **93 %** of ATP synthase's protons; all twelve uncostered proton-movers run the dissipative way; **no free path exists at any length**
- ▶ Then we repeated one of our own structural findings at **their published posterior** — 100 models, their code, their parameters: changing which constraint binds widens the top of the TPC from **1.5 to 7.3 °C in 93 %** of their models
- ▶ **But CT_{max} also moves (4.6 °C).** In *our* model it did not. **The asymmetry is not the clean thing we thought it was**
- ▶ Register: an audit applied to the reference implementation — not a critique of a paper

The asymmetry at Li et al.'s own posterior

Left: the top of the curve widens from ~ 1.5 to ~ 7.3 °C. **Right:** T_{opt} moves further than CT_{max} in 92 % of their models — but CT_{max} is **not** immune.

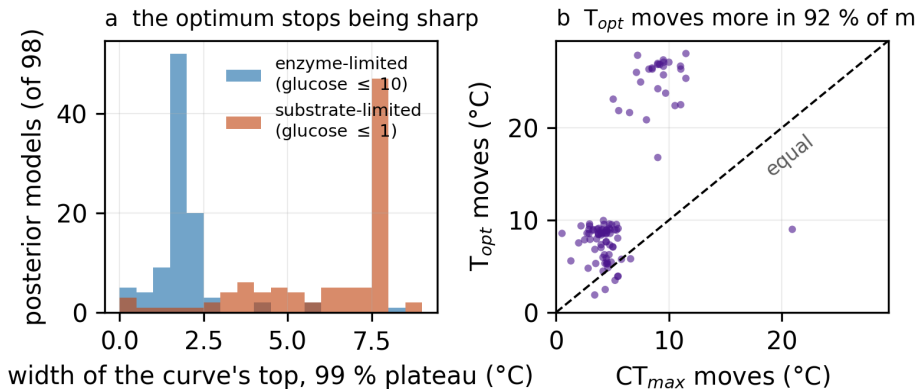


Figure 13: reports/Y2_regime_posterior

The awkward one: our model needs a -4 K stability shift

- ▶ Every fit of the *E. coli* model wants melting temperatures **3–4.5 K below a measured meltome**¹⁰
- ▶ Two readings, and we screened them: **biology** (*in vitro* T_m is not *in vivo* inactivation) or **our parameterisation** (16 global levers where Li et al.¹ had 2 292)
- ▶ **The shift is not uniquely required** — an equally good, living fit exists at zero shift, for 0.08 log-likelihood units
- ▶ **But catalysis is not what pays for it.** The compensator is `tm_scale`, the *spread* of melting temperatures. Both solutions put the least-stable few per cent of enzymes **4–6 °C below measurement**
- ▶ So the **number** is a parameterisation artefact; the **contradiction with the meltome's low tail** is not. Per-enzyme $k_{cat}(T)$ would not fix it — and the predictor we would have used has no usable optimum for 97 % of these enzymes

...and the same degeneracy in two numbers

- ▶ **Pin one melting-temperature lever and the other compensates: 0.08 log-likelihood units.** Effectively free
- ▶ **Pin both to the measurement and it costs 7.6 units.** The organism still grows — 1.66 h^{-1} — but the fit is pressed against two prior bounds
- ▶ **So the two levers are constrained together and neither alone.** That is the same $dT_m/\text{tm_scale}$ pair the posterior run's own geometry found, in units anyone can read
- ▶ One method was an optimiser profiling the likelihood; the other was a sampler's live points collapsing onto a correlation. **They share no machinery**

Established results and pending confirmation

- ▶ **Settled.** One shared codebase with seven organism models on it, under the regression suite. The mechanisms: measured proteome sectors, overflow emerging from a carbon cap, the ETC area budget. The audits — including the reference implementation passing ours
- ▶ **Settled, and uncomfortable.** No interval from the older *E. coli* fits is available. The likelihood surface is structurally rough. The meltome's low tail is contradicted by every fit
- ▶ **Diagnosed, not yet solved.** Five independent runs have now met their stopping rules across two versions of the likelihood, and **no two of them agree**. The scoring defect that rewarded infeasible models is fixed and measured; the remaining obstacle is that the likelihood is not reproducible in a worker pool, which is now measured too, with a remedy costed
- ▶ **Needs measurement — not more modelling.** Proteome allocation **above 37 °C on glucose** — the measurement exists, its *range* does not, and that is where CT_{max} is set; maintenance vs temperature from a low-dilution chemostat (ngam_* are borrowed); *in vivo* stability for the enzymes that set the thermal limit

Has it ever predicted anything it was not fitted to?

- ▶ **Once, and it half-worked.** The *uncalibrated* model was asked to predict the exact-strain Van Derlinden curve with **nothing fitted to growth**: it tracked the **shape** (T_{opt} , E_a) and under-predicted the **peak ~2.3-fold** (1.04 against 2.40 h⁻¹). Shape held; magnitude did not
- ▶ **And then that curve was consumed by the calibration**, so the *calibrated* model has no holdout left. Cooper et al.'s evolved *E. coli* lines¹¹ are the obvious next one

The next controlled comparison: configuration D on M9

- ▶ **Change the medium, keep the configuration.** M9 plus glucose is fully defined. NLDM is a defined model recipe standing in for the experiments' undefined R2A medium
- ▶ If the degeneracy and low-growth mass recur on M9, the explanation extends beyond NLDM. If they change, the medium is implicated
- ▶ **Before fitting:** resolve infeasibility scoring, check oxygen-flux tie-breaking on M9 and whether D's carbon cap binds. Keep the excluded M9 control series explicit
- ▶ A more restrictive medium could increase infeasibility. This comparison follows the likelihood diagnosis

Model evidence asks which mechanism the data support

- ▶ **D: a total-carbon cap** produces overflow
- ▶ **E: a respiratory membrane-area budget** limits respiratory capacity
- ▶ **F: that area budget with non-electrogenic bd-II** changes respiratory coupling
- ▶ These are **competing mechanisms**. Compare their log evidences on the **same medium and observations**, with a common scoring rule and declared comparable priors
- ▶ Nested sampling returns evidence with the posterior. Under the **revised** likelihood, D on NLDM gives -27.707 ± 0.110 and -28.047 ± 0.100 on two independent seeds — which **do not agree** within their combined error
- ▶ **So there is no D value to compare against yet.** The comparison needs a reproducible number first; that is one remedy away, not one run away
- ▶ A further caution for E and F: their oxygen flux sits on a *face* far more often than D's, so the reproducibility obstacle is **an order of magnitude larger** there

What happens next

- ▶ **Decide the determinism remedy** — a fresh model per evaluation is the only exact one, at $3.2\times$ cost. Everything below inherits it.
- ▶ **Then one run under that remedy**, to see whether the seeds still disagree. That is the cheap test of whether a second cause is real.
- ▶ Test D on fully defined M9.
- ▶ Compare D/E/F evidences on one medium.
- ▶ Measure glucose proteome allocation above 37 °C.

What is handed over

- ▶ **A working framework.** Seven organism models on one codebase, two regression gates passing at every commit (79/79 and 60/60), the mechanisms implemented as options that are off until a strain turns them on
- ▶ **A likelihood that is right, and an inference that is not.** The scoring rule now treats infeasibility correctly and every instrument that established it is committed and re-runnable
- ▶ **A named obstacle with a costed remedy**, rather than an unexplained failure — and a second cause still open, stated as open
- ▶ **Every self-correction on the record:** the multimodality reading withdrawn, a parameter found never to be read, a “96%” corrected to 16.5%, a remedy measured and found worse than doing nothing
- ▶ **The decisions that are the group’s**, gathered in one list with owners, in docs/HANDOVER.md

Three questions for the group

1. **Which respiratory mechanism should the data distinguish?** A carbon cap, a membrane-area limit, or altered respiratory coupling? What independent measurement would make the evidence comparison convincing?
2. **How should infeasible predictions be scored?** We can identify the exemption. What observation model would make a model with no feasible flux solution appropriately unlikely?
3. **What would count as a strong external test?** Predicting thermal-curve shifts after 20,000 generations¹¹, or another held-out experiment? Glucose proteome measurements above 37 °C would constrain the uncertain allocation layer

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